

Health data information and management System (HDIMS) mobile application

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ABSTRACT

An essential component of effective healthcare delivery is health record management. Although, conventional paper-based systems have long been the standard, they have a number of drawbacks, including inaccuracy, inefficiency, and a lack of accessibility. The adoption of a web-based health record management system proved to be a viable replacement for the paper-based system for handling medical records. However, there were certain obstacles along the way, including concerns about data security and privacy, data loss during the shift from paper to the web-based system, a lack of inclusivity, unauthorised access to patient records, and limited access to patient records because of inadequate internet access and facilities in Nigeria. This dissertation therefore reviewed the observable limitations of the existing system and proposed the development of a mobile patient's health record management system using mobile application development framework. The system was developed using Android Studio integrated development environment (IDE) and the Java programming language, with SQLite database and rapid application development methodology (RAD) which is an Agile method for enhanced program development especially for mobile application development. The new system outperformed the existing web-based hospital record management system in terms of speed, efficiency, robustness portability, data security and integrity as it enables both patients and doctors to access and book appointments remotely without having to go physically to the hospital saving man-hour and boosting the morale of both staff and patients.

1. INTRODUCTION

A good healthcare system depends on the effective management of healthcare information, ensuring it is accessible, accurate, and available to healthcare providers. Proper management of information in healthcare is vital for the growth and efficiency of healthcare organizations, as it

can mean the difference between life and death. Healthcare institutions face the challenge of accurately maintaining patient records, which is crucial for providing effective care (Shimishi and Mupala, 2019).

Advances in healthcare are rapid, driven by the need to manage increasing health challenges. Technology has facilitated the

storage, search, and access of patient records, reducing reliance on physical filing systems. However, Nigeria has been slow to adopt these technological advances, hindered by factors such as inadequate funding, training, and a perception that technology is only useful in developed countries. Many Nigerian healthcare facilities are understaffed and overcrowded, leading to delays in patient care. Patient conditions can deteriorate while searching for records, emphasizing the need for systems that store essential patient data for timely medical responses.

Since the 1990s, the healthcare sector has integrated information and communication technology (ICT) to improve service quality in areas like patient safety, efficiency, and satisfaction. The goal of ICT in healthcare is to enhance access, quality, and effectiveness for all citizens. ICT has proven to be a key tool in facilitating electronic communication, processing, and information transmission, ultimately improving health outcomes (Bates and Gawande, 2002). The role of ICT in improving healthcare and increasing the availability of health information is vital, especially in developing nations where it can enhance service quality and reduce costs (Brandrud et al., 2011).

Patient health information management systems have the potential to monitor health records and treatments over time, providing insights into the best approaches for patient care and diagnosis (Gomes and Romão, 2020). insights into their health status, whether for chronic or acute conditions.

1.6 Factors to be considered for using Mobile Healthcare Application

There are certain factors that should be considered when developing and utilising mobile healthcare apps for patient and personal healthcare in any health setting

of the country. According to Dike and Kabir (2020), these factors could include:

- i. proportion of people with a mobile phone in a population in the region
- ii. internet accessibility,
- iii climatic/environmental conditions, data security and,
- iv literacy level of the people.

When the population of people who own and use mobile devices is small, it makes it challenging to implement them for widespread use in healthcare delivery in the region. Poor network and internet services impede the widespread use of mobile healthcare applications (mHealth apps) for care delivery, and reliance on them may worsen people's healthcare status.

2. METHODOLOGY

This research adopted the rapid application development methodology (RAD). This methodology is an information systems development procedure with a lifecycle designed to render speedy development and increased-quality reports. It is designed to have the maximum advantage of strong development software that has excellent quality, risk control and project delivery on time within the given frame of time budget. This can also be seen as an approach to constructing computer systems that put together computer-assisted software engineering tools and techniques, user-driven prototyping, and stringent project delivery time limits into a potent, tested, trusted formula for the best quality and productivity. This development drastically uplifts the importance of finished systems while decreasing the minutes it uses to construct them. This is achieved by accustoming sets of trusted application development tools,

around a well-defined method. It is a process from which the development cycle of an application is expedited. Finally, RAD is a method of software development with an object-oriented approach to system development (Dennis, 2012).

This Rapid Application Development (RAD) methodology is flexible and adaptable to changes. The following are some of the rationale for the choice of the methodology for the development of a mobile patient's record management system;

- i. reduction of the overall project risk,
- ii. it is easier to transfer deliverables as scripts,
- iii. high level abstractions and intermediate codes are used,
- iv. it enables code generators, code reuse and reduces manual coding,
- v. it is prototyping in nature; there is a possibility of a lesser defect,
- vi. every phrase in the Methodology delivers highest priority functionality to client with less people and increased productivity in a short time

Figure 1 Architecture of the Proposed Mobile Patient's Record Management System

record management system.

Description of the proposed Systems Architecture

The proposed system architecture in figure 1 comprises of three main modules which includes;

- i. the login/ account creation module,
- ii. patients mobile platform,
- iii. admin/ record management

module and

- iv. physician report modules.

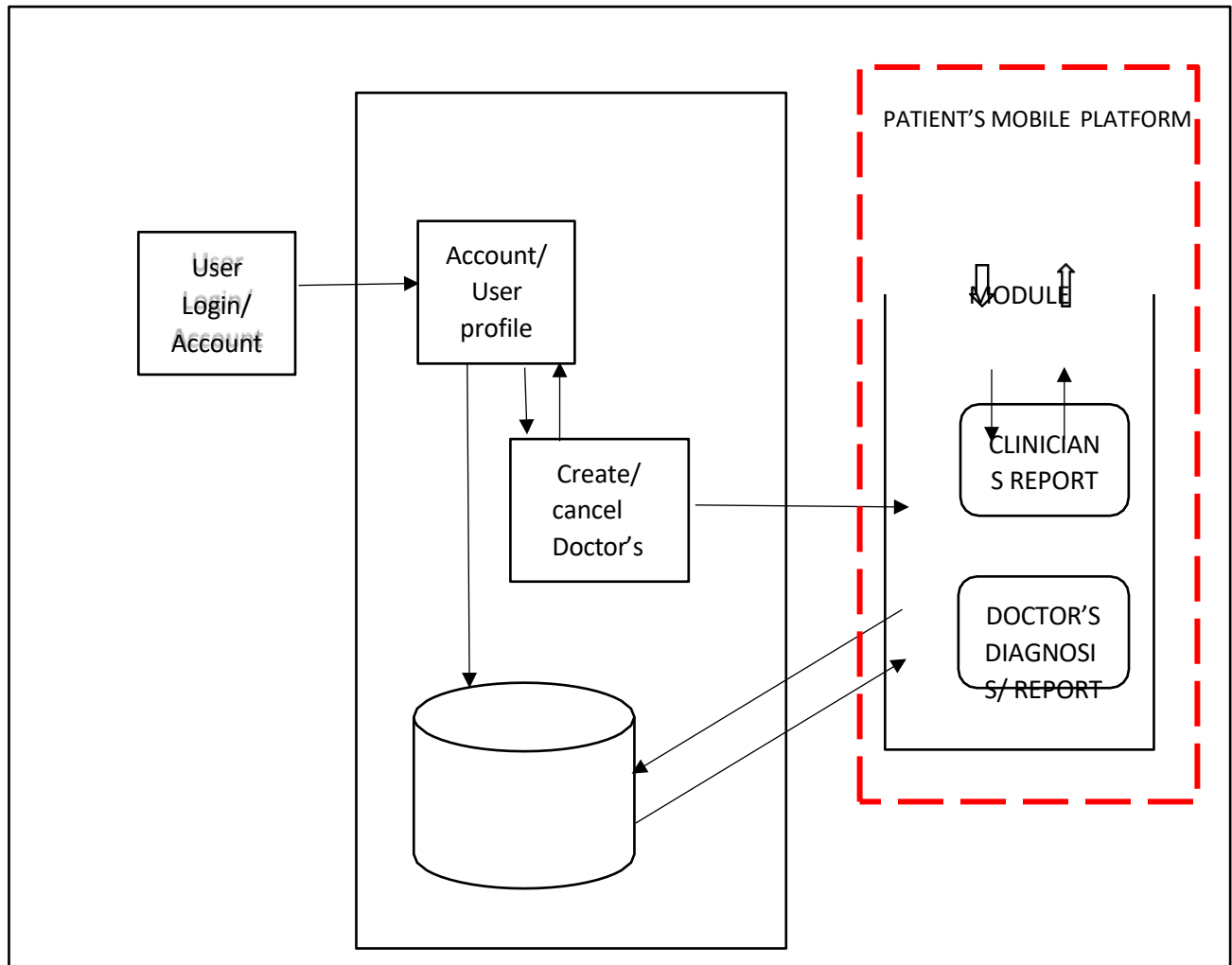
At the login/ account creation module, the user/ patient after getting or downloading the mobile patient management application signs up/ create a user account with a distinct username and password that will allow him gain access into the application to update his profile. If the user does not have an existing account he cannot use the application. Once the account has been created the user automatically has an electronic medical record folder created for him/her in the given healthcare institution/ hospital. Furthermore, the user also known as the patient can book or create a doctor's appointment which will be stored on the systems database to be seen only and updated by the admin and doctors respectively. It is important to note that from the appointment creation module the user can also cancel his appointment if he cannot attend. The doctors from their own end gain access to all the appointment, attends to the patients and the saves their report in the systems' database for future use. To this end, when a patient has a repeat visit or has an emergency, it is easy to access their medical history/ records by a click and medical attention given to them. indicates the different database tables in the proposed system.

2.7 Security Design of the Proposed System

The security design of the proposed system includes the data security and privacy concern of the system. This ensure that only authorized users have access to the system to reduce the risk of data loss, theft and human error. The security ranges for the login restriction to the data type deployed for the system. The security design includes the processes put in place to manage both human and software issues that may arise in the course of the design, development and implementation of the system.

3. RESULTS ANALYSIS AND DISCUSSIONS

This dissertation is implemented using mobile application development framework

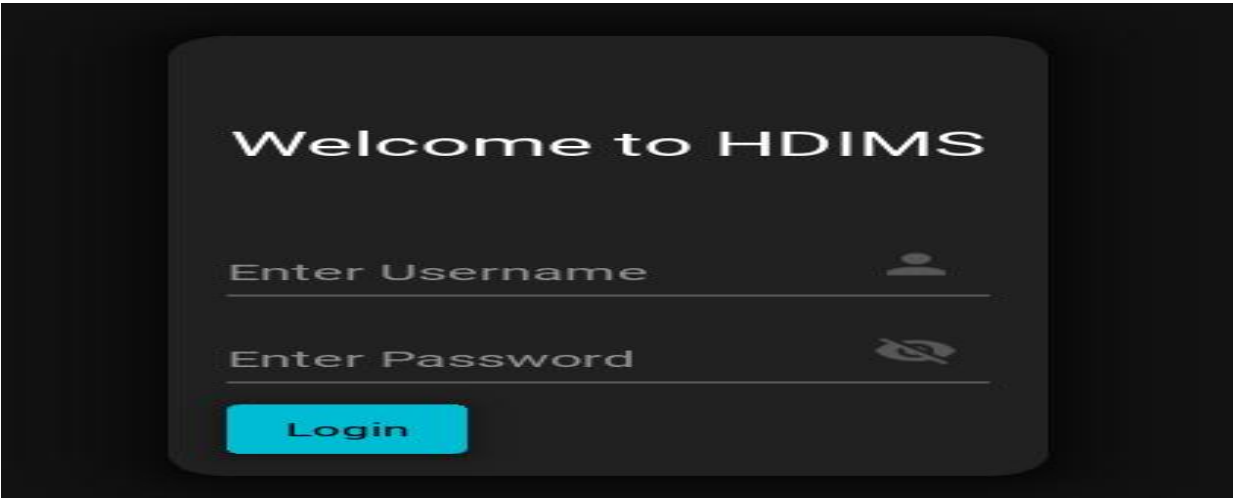


with an integrated development environment (IDE) that as the application is designed to allow the user perform the basic operation as it manly an android base platform that is designed to support provision of healthcare delivery services, and allow the user to register, book appointment and check their appointment status on the healthcare facility or hospital they are registered with. It also gives the medical practitioners the opportunity to see their proposed patients for the day and also reschedule if they cannot attend to them. As stated earlier the project is an android base application and the implementation programming language is Java programming language on android

studio integrated development environment (IDE).

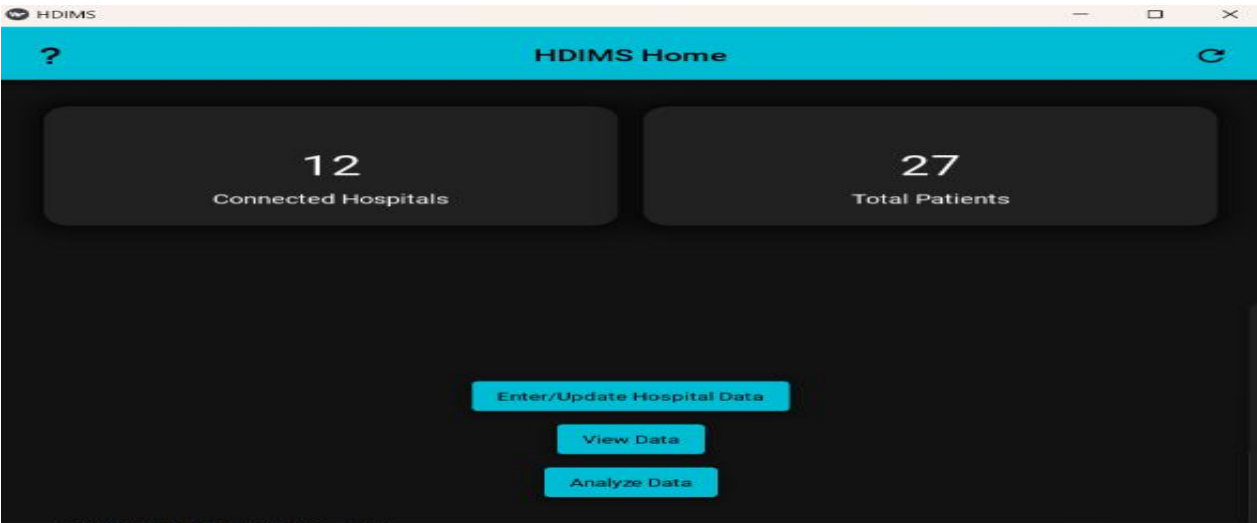
Login Form: This is where users will sign in to use the application. It is important to note that before any user can use the system, that user must have registered/ signedup on the application to be able to use the system. The login serves as a security check for unauthorized access into the system as it works as user authentication module of the system. The login form Figure-6

Registration Form: The registration module allows users to enter personal details after downloading the app to register with the hospital system. The data is stored in the hospital's database,



A login form titled "Welcome to HDIMS" on a dark background. It features two input fields: "Enter Username" with a person icon and "Enter Password" with a lock icon. A red "Login" button is positioned below the password field.

Figure 6 Login Form



The HDIMS Home dashboard interface. It includes a top navigation bar with a question mark icon, the text "HDIMS Home", and a refresh icon. Below the navigation bar, there are two summary cards: "12 Connected Hospitals" and "27 Total Patients". At the bottom, there are three red buttons: "Enter/Update Hospital Data", "View Data", and "Analyze Data".

Figure 7 Home Info

Figure 6 Login Form

The screenshot shows a mobile application window titled 'HDIMS' with a blue header bar labeled 'Hospital Data Entry'. Below the header, there are four input fields, each with a placeholder text and a corresponding icon: 'Enter Hospital Name' (building icon), 'Enter Patient Name' (person icon), 'Enter Disease Name' (sun icon), and 'Enter Doctor Name' (person icon). A blue 'Submit Data' button is located at the bottom center of the form.

Figure 7 Home Info

The screenshot shows the same mobile application window as Figure 6, but with the input fields filled with data: 'Sharma' for Hospital Name, 'raja' for Patient Name, 'tb' for Disease Name, and 'VASU' for Doctor Name. The 'Submit Data' button remains at the bottom.

Figure 8 Patient Entry Form

Doctors' Module: This module is the part of the mobile application where the doctor/ physician logs in to attend to patients and also review what has already been done. In this module the doctor accesses the list of patients that have made appointment for the day and also attends to them. This module also have sub-module that gives the doctor/ physician access to the patients record so that as they finish attending to a particular patient, they save and the patient's record is updated for

Figure 9 Patients details

future purposes thereby keeping track of the patients health record. This electronic record is updated from time to time as the patient visits the healthcare facility for on health related purpose or the other and also as they use their mobile applications to book and make

request from the system. The doctor's modules in figure 8 and 9 is used to view and manages appointment.

Hospital Name	Patient Name	Disease Name	Doctor Name
aiims	aman	tb	bharat
lelavati	harsh	viral	ram
bharti	adarsh	viral	sham
poona	manish	viral	sita
sasun	harshita	food poison	arjun
sanchati	manisha	jaundice	ram
command	pratamesh	food poison	bhim
leharanir	nirula	aids	navan

Figure 10 Patients Module

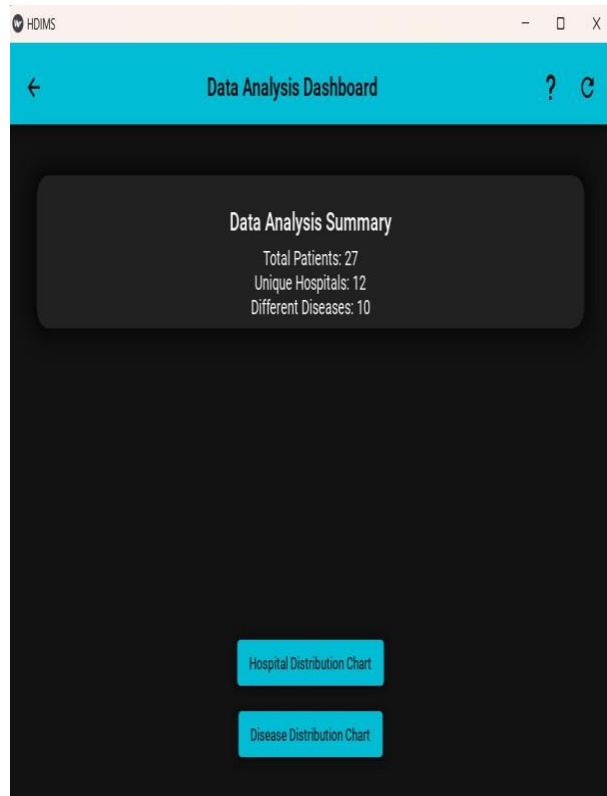


Figure 11 Data Analysis Summary

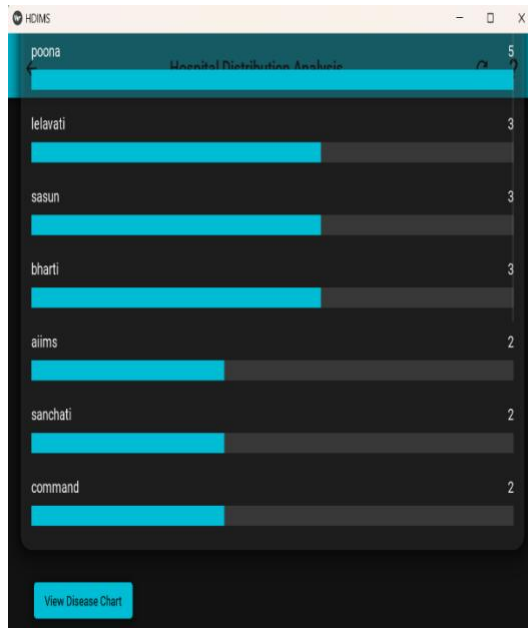


Figure 12 Disease Chart



Figure 13 Hospital Chart

Setting up an appointment with the doctor as indicated in Figure 11, the patient will have the opportunity to book an appointment with a doctor of his or her choice from the list of available doctors in the hospital. Also Figure 12 enables the

3. Accessibility: Enhanced by 60% (30% to 90%)
4. Cost: Reduced by 60% (80% to 20%)
5. Access Time: Improved by 80% (50% to 90%)

Table 4 Empirical Analysis Result of the Proposed System

S/N	Functionalities	Existing System	New System	Improvement	Statistical Significance
1	Portability	60%	90%	30%	0.01
2	User Experiences	40%	90%	50%	0.005
3	Accessibility	30%	90%	60%	0.001
4	Cost	80%	20%	60%	0.01
5	Access Time	50%	90%	40%	0.05
6	Appointment Booking Time	30%	90%	60%	0.001
7	Data Security	40%	90%	50%	0.005

patient to view their appointments and know when they have an appointment with the doctors to visit the hospital.
Empirical Analysis of the Result

The proposed mobile healthcare record management system outstands the existing system in all functional areas:

1. Portability: Increased by 30% (60% to 90%)
2. Ease/User Experience: Improved by 50% (40% to 90%)

Comparative Analysis of the Existing and Proposed System

After an exhaustive comparative analysis of the functionality of the existing system by Yahaya et al (2019) adopted in the

6. Appointment Booking Time: Reduced by 60% (30% to 90%)
7. Data Security: Enhanced by 50% (40% to 90%)

The proposed system demonstrates significant improvements in user experience, accessibility, and data security, while reducing costs and appointment booking times. The development of the UNIBEN Teaching Hospital (UBTH) hospital information management system and the proposed mobile healthcare record management system, the new systems outperformed the existing system in the following area as indicated in Table 5

Table 5 Comparison of the Existing

S/N o	Functionalities	Existing system	Proposed/ New System
1	Portability	Portable	More Portable
2	Ease/ User Experience	Poor	High
3	Accessibility	Low	High
4	Cost	High	Low
5	Access time	30 Minutes	60 seconds
6	Appointment booking time	1 hour	5 Minutes
7	Data Security	Low	High

One of the major setbacks of the existing system that the new mobile healthcare information management system overcame is the issue of time for booking and checking patients for doctor's appointment.

Result Tested Analysis of the Proposed System

1. Healthcare Professionals' Feedback (Doctors, Nurses, Admin Staff) •

2. **Improved Accessibility:** *"With this system, I can access patient records instantly from my mobile device, reducing waiting time by at least 50%."* – Dr. Kare, General Practitioner

3. **Reduced Errors:** *"Prescription errors have significantly decreased since we started using the system, as it flags potential drug interactions in real-time."* – Nurse Chioma, Senior Nurse

4. **Ease of Use:** *"The interface is intuitive, making it easier for new staff to adapt without extensive training."* –

Hospital IT Manager

manually, but now the doctor has my records available instantly." – Fatima, 30-year-old mother • **Security Concerns:** *"I appreciate the added security features like biometric login, which make me feel my health records are safe."* – Chinedu, a 50-year-old patient

5. Administrative and Operational Insights •

Cost Reduction: *"Printing and storing paper records was a major expense. This system has cut those costs by 60%."* – Hospital Administrator

• **Efficiency in Record Keeping:** *"With the automated backup and cloud synchronization, we no longer worry about data loss."* – IT Specialist

• **Interoperability:** *"The ability to integrate with laboratory and pharmacy systems has streamlined operations significantly."* – Medical Records Officer

6. Challenges and Areas for Improvement

• **Network Dependency:** *"The system relies heavily on internet connectivity, which can be a challenge in remote areas."* – Clinic Staff in Rural Area • **Initial Learning Curve:** *"Some older staff members needed additional training to transition from paper-based records to the digital system."* – Training Coordinator

4. CONCLUSION

The importance of an enhanced healthcare record management system cannot be overemphasized in a technologically evolving world like ours as these changes within split seconds and human need as desire for healthcare service delivery increases so also the need for

proper health record management system. The miniaturization of the electronic health record management information system to mobile application has served as a tool for enhance healthcare delivery as patients can remotely and view their health record and make appointment to see the doctor without necessarily visiting the hospital or healthcare institution to do so. This has not only enhanced the record management system of the healthcare sector; it also has improved the level of their clientel and boost the morale of the staff. This has also improved accessibility, reduce booking time, allocation of doctors, recuded cost and enhanced the user experience (UX).

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